

# Web Application Security Report

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**Internship Domain:** Cybersecurity

**Task:** Web Application Security Basics

**Target Application:** DVWA (Damn Vulnerable Web Application)

## 1. Overview

This task focuses on understanding basic web application vulnerabilities using a locally hosted intentionally vulnerable application. The assessment was performed to learn how common attacks such as Cross-Site Scripting (XSS) and SQL Injection are identified and analyzed using cybersecurity tools.

## 2. Tools Utilized

- DVWA (Damn Vulnerable Web Application)
- XAMPP (Apache + MySQL)
- Burp Suite Community Edition

## 3. Target Environment

**Target:** DVWA

**URL:** http://localhost/dvwa

**Platform:** Windows using XAMPP

**Security Level:** Low

## 4. Approach Followed

### Step 1 – Environment Setup

DVWA was installed and configured on a local Windows machine using XAMPP.

### Step 2 – Vulnerability Testing

Cross-Site Scripting (XSS) and SQL Injection vulnerabilities were tested using crafted payloads.

### Step 3 – Traffic Analysis

Burp Suite was used to intercept and inspect HTTP requests and responses during testing.

## 5. Vulnerability 1 – Cross-Site Scripting (XSS)

**Module:** XSS (Reflected)

**Payload Used:** <script>alert('XSS')</script>

**Result:** Alert popup appeared in the browser, confirming the vulnerability.

## 6. Vulnerability 2 – SQL Injection

**Module:** SQL Injection

**Payload Used:** 1' OR '1'='1

**Result:** Database records were exposed, confirming SQL Injection vulnerability.

## 7. Identified Security Issues

- Lack of input sanitization and validation
- Direct SQL query construction using user input
- JavaScript injection possible through input fields
- Missing Content Security Policy (CSP) headers
- Exposure of database errors and sensitive information

## 8. Risk Evaluation

**Critical Risk:** SQL Injection – full database access possible

**High Risk:** Cross-Site Scripting – session hijacking and data theft possible

**Medium Risk:** Missing security headers and weak input validation

## 9. Recommendations

- Use prepared statements and parameterized queries
- Sanitize and encode all user inputs
- Implement Content Security Policy (CSP) headers
- Validate inputs on both client and server side
- Hide raw database errors from users
- Use a Web Application Firewall (WAF)

## 10. How to Perform This Task

**Step 1:** Install XAMPP and start Apache and MySQL services.

**Step 2:** Download and configure DVWA inside the htdocs folder.

**Step 3:** Open DVWA in browser using:  
`http://localhost/dvwa`

**Step 4:** Set DVWA security level to Low.

**Step 5:** Test XSS vulnerability using payload:  
`<script>alert('XSS')</script>`

**Step 6:** Test SQL Injection using payload:  
`1' OR '1'='1`

**Step 7:** Use Burp Suite to intercept and analyze HTTP traffic.

## 11. Final Result

The web application security assessment was successfully completed using DVWA and Burp Suite. The testing identified major vulnerabilities including Cross-Site Scripting (XSS) and SQL Injection. Appropriate mitigation techniques and security recommendations were provided to improve application security.

## 12. Conclusion

This task demonstrated how common web application vulnerabilities can be exploited when secure coding practices are not followed. Proper validation, secure query handling, and regular security testing are essential to protect web applications from attacks and data breaches.

## 13. Viva Questions and Answers

### Q1. What is DVWA?

DVWA is an intentionally vulnerable web application used for cybersecurity practice.

### Q2. What is XSS?

Cross-Site Scripting is a vulnerability that allows attackers to inject malicious JavaScript code into web pages.

### Q3. What is SQL Injection?

SQL Injection is an attack where malicious SQL queries are inserted through user input fields.

### Q4. What is Burp Suite?

Burp Suite is a web security testing tool used to intercept and analyze HTTP traffic.

## 14. References

- OWASP Web Security Guidelines
- DVWA Documentation
- Burp Suite Documentation